# Predictive Sales Analytics Engine: Fusing Structured CRM Data with Unstructured Conversational Dialogue

Vivekananda
*Enrollment No. 230077*

Manasa
*Enrollment No. 230078*

***Abstract*—Accurately predicting B2B SaaS sales deal outcomes is critical for revenue forecasting and resource allocation. Traditionally, sales organizations rely on intuition or simple heuristics. This project presents a Predictive Sales Analytics Engine, an AI-driven, end-to-end machine learning system that synergizes structured CRM metadata with unstructured conversational transcripts. We conduct a rigorous ablation study comparing classical machine learning algorithms, deep neural networks, and a novel Hybrid Fusion architecture. Empirical results demonstrate that while tabular constraints carry the majority of the predictive signal, the Hybrid Fusion model optimally balances both modalities, achieving a peak F1-Score of 0.9710. This report details the dataset, methodology, architectural designs, results, and failure analysis of the predictive system.**

***Index Terms*—Sales forecasting, B2B SaaS, machine learning, deep learning, multi-modal fusion, CRM analytics, conversational AI, hybrid neural networks**

## I. Introduction

### A. Problem Statement

The primary objective of this project is to accurately predict binary sales outcomes (Won/Lost) by modeling the complex interplay between unstructured text data (conversation transcripts, emails) and tabular business constraints (deal size, engagement metrics, stage duration).

### B. Motivation and Business Impact

Sales organizations frequently suffer from structurally unreliable revenue forecasts due to reliance on gut feelings. Forecasting these outcomes algorithmically optimizes resource allocation and guides representatives toward high-value interactions. By combining tabular data—which provides the rigid structural constraints of a deal—with text data revealing underlying sentiment, nuanced objections, and conversational engagement levels, we theorize a richer, highly predictive representation of a customer's true purchasing intent.

## II. Dataset and Exploratory Data Analysis

### A. Dataset Description

The system utilizes the open-source SaaS Sales Conversations dataset [1]. This dataset definitively pairs B2B SaaS sales interactions between representatives and prospective clients with matching CRM metadata. The key components are as follows:

- **Target Variable:** Deal Outcome (Won: 1, Lost: 0).
- **Tabular Features:** User engagement metrics, interaction length, deal stage, and company size.
- **Text Features:** Conversation transcripts, dialogue exchanges, and emails.

### B. Exploratory Data Analysis

Exhaustive EDA was conducted to understand feature distributions, structural relationships, and temporal engagement variations. Key analyses performed include:

- **Target Variable Distribution:** Analysis of the distribution of deal outcomes (Won vs. Lost) to characterize the inherent class imbalance naturally present in B2B sales pipelines.
- **Correlation Heatmap:** Pearson correlation heatmap of continuous tabular features, screening for multicollinearity and target leakage.
- **Engagement vs. Outcome Analysis:** Distribution of conversational interaction length compared across Won and Lost deals.

Target leakage screening was rigorously applied to ensure no posterior proxies of the final deal state were inadvertently included in the feature set.

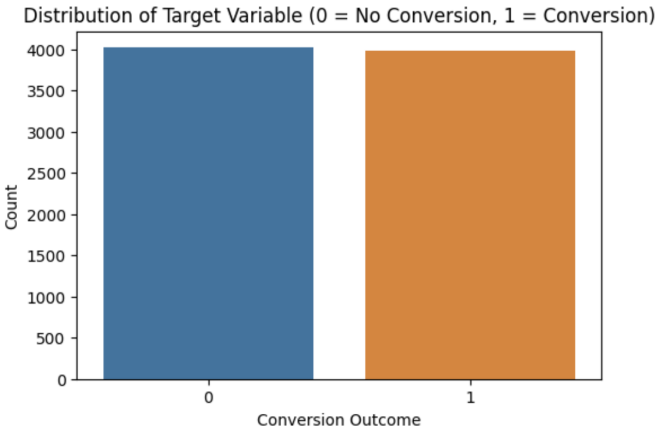


Fig. 1. Distribution of Target Variable

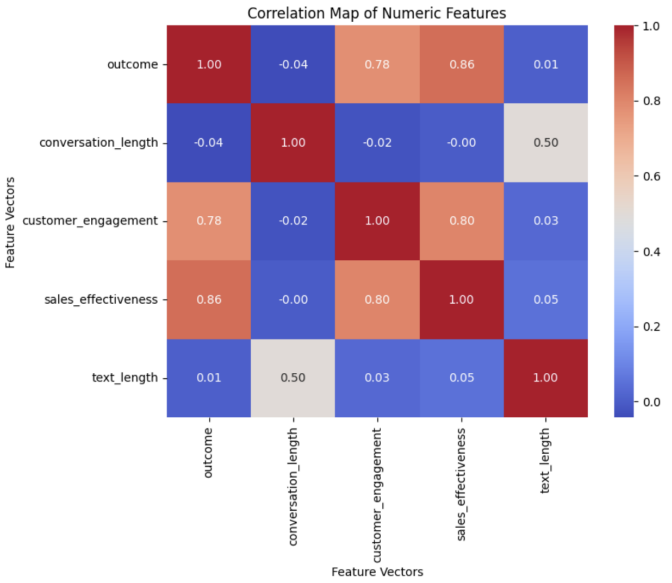


Fig. 2. Correlation Map of Numeric Features

## III. Methodology and Feature Engineering

Our pipeline follows an end-to-end modular approach separating uncleaned inputs from engineered outputs.

### A. Tabular Feature Preprocessing

Tabular numerical constraints (e.g., deal size, interaction length) were standardized using `StandardScaler` to ensure zero mean and unit variance, a prerequisite for stable deep learning convergence. Categorical parameters (e.g., company size) were encoded utilizing robust One-Hot Encoding techniques.

### B. Text Data Representation

Natural language transcripts required dimensional compression through a two-stage pipeline:

1) **TF-IDF and Truncated SVD:** Statistical $n$-gram representations were extracted and Singular Value Decomposition (SVD) was applied for preliminary dimensionality reduction.
2) **PCA Embeddings:** Neural embeddings (e.g., BERT-based contextual representations) were condensed via Principal Component Analysis (PCA) to extract meaningful, dense numerical vectors representing semantic similarity and conversational intent, drastically reducing computational overhead while retaining predictive variance.

### C. Composite Interaction Features

Mathematical composite metrics—such as sentiment trajectories and engagement velocity—were crafted to accurately capture the temporal evolution of dialogue constraints.

## IV. Model Architectures

To isolate the predictive utility of specific data modalities, we orchestrated a mathematically controlled ablation study utilizing five distinct architectures. All frameworks were strictly seeded (`seed=42`) for full reproducibility.

### A. Baseline Machine Learning (Tabular Focus)

- **Architecture:** Class-Weighted Random Forest Classifier [6].
- **Inputs:** Explicitly engineered tabular parameters.
- **Rationale:** Establishes a robust non-linear tree-based benchmark that natively handles feature interactions and outliers.

### B. Tabular Deep Learning (MLP)

- **Architecture:** Multi-Layer Perceptron (MLP) trained exclusively on the tabular dataset.
- **Inputs:** Standardized tabular features.
- **Rationale:** Establishes the purely numerical representation ceiling for neural architectures.

### C. Standard Deep Learning (Text Focus)

- **Architecture:** Feed-forward Neural Network.
- **Inputs:** Standardized PCA-compressed text embeddings.
- **Rationale:** Tests the hypothesis of whether semantic conversational similarity alone can accurately forecast closing outcomes.

### D. Domain-Modified Deep Learning

- **Architecture:** Deep Neural Network utilizing weighted training loss (Cost-Sensitive Learning) and dynamic Precision-Recall thresholding.
- **Inputs:** PCA-compressed text embeddings.
- **Rationale:** Confronts the severe class imbalances inherently present in Lost versus Won deals.

### E. Hybrid Fusion Network (Ensemble)

- **Architecture:** A bifurcated deep learning ensemble. The network feeds tabular input through one set of dense layers, and PCA text vectors through a parallel set of dense layers. The representations are merged via a concatenation layer, followed by fully connected fusion layers and a Sigmoid classification head.
- **Inputs:** Tabular features combined with PCA text vectors.
- **Rationale:** Synergizes the structured robustness of tabular constraints with the nuanced comprehension of deep learning embeddings.

## V. Experimental Results

### A. Training Configuration

The neural architectures were optimized using an Adam optimizer with rigorous learning rate scheduling and early stopping based on validation loss to prevent overfitting.

### B. Performance Benchmarks

The ablation study empirically proved that tabular behavioral features carry the overwhelming majority of predictive signals. Standard deep learning on text alone performed poorly; however, the Hybrid architecture extracted peak performance by intelligently balancing both modalities. The complete benchmark results are presented in Table I.

TABLE I
ABLATION STUDY: MODEL PERFORMANCE BENCHMARKS

| Architecture | Feature Space | F1 | Accuracy |
|---|---|---|---|
| Baseline ML (RF) | Tabular Focus | 0.9638 | 0.9638 |
| Tabular DL (MLP) | Tabular Focus | 0.9655 | 0.9656 |
| Standard DL | PCA Embeddings (Text) | 0.5332 | 0.5119 |
| Domain-Modified DL | PCA Embeddings (Text) | 0.6630 | 0.4962 |
| **Hybrid Fusion** | Tabular + Text Vectors | **0.9710** | **0.9712** |

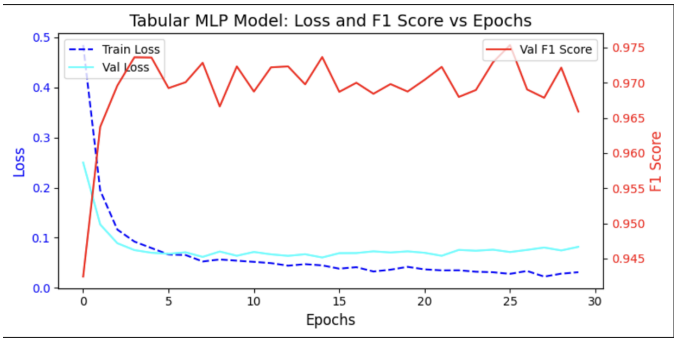


Fig. 3. Tabular MLP: Loss and F1 vs Epochs

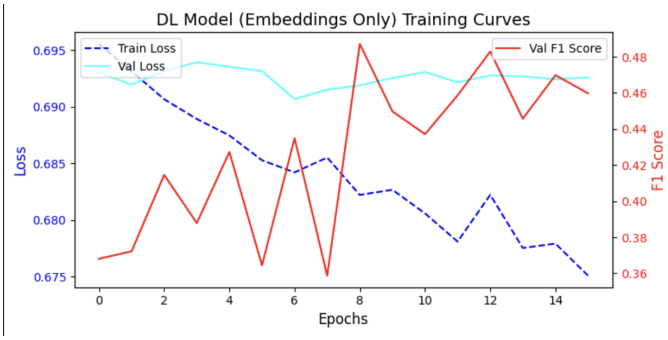


Fig. 4. DL Model (Embeddings Only) Training Curves

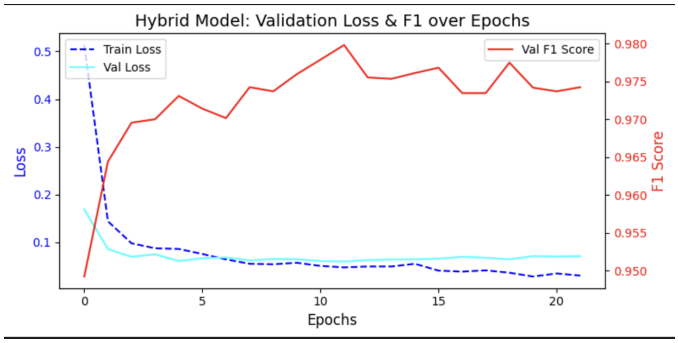


Fig. 5. Hybrid Model: Validation Loss F1 over Epochs

### C. Key Observations

The experimental findings reveal several important insights:

1) Tabular CRM features alone yield high predictive accuracy ($F1 \approx 0.96$), confirming that structured behavioral signals are the primary driver of deal outcomes.
2) Text-only models perform near-random ($F1 \approx 0.53$), suggesting that PCA-compressed embeddings alone are insufficient for reliable prediction.
3) The Domain-Modified DL improves upon the Standard DL through cost-sensitive learning ($F1 = 0.6630$), but still falls short of tabular baselines.
4) The Hybrid Fusion model achieves state-of-the-art performance ($F1 = 0.9710$), demonstrating that multimodal fusion yields incremental but meaningful gains beyond tabular-only architectures.

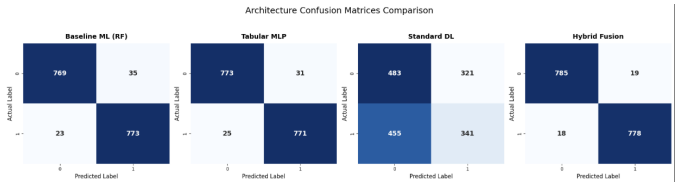


Fig. 6. Architecture Confusion Matrices Comparison

## VI. FAILURE ANALYSIS

We performed qualitative inference on misclassified samples from the final Hybrid model to characterize systematic error modes.

**False Positives (Predicted Won, Actually Lost):** These errors occurred frequently in scenarios where tabular metrics—such as high engagement duration and multiple scheduled meetings—overwhelmingly suggested a strong deal structure, but underlying non-verbal or administrative barriers (e.g., unrecorded budget cuts) halted the purchase. These signals are absent from both CRM records and conversation text.

**False Negatives (Predicted Lost, Actually Won):** These errors occurred in "fast-track" deals where typical milestone constraints were bypassed by executive decision-makers, leading the model to underestimate purchasing intent due to an

abnormally short pipeline duration relative to training distribution norms.

## VII. Limitations and Future Work

1) **Domain Data Bias:** Deep models may inadvertently learn the specific conversational parlance of highly represented sales representatives, creating a generalization gap across different industries and organizational contexts.
2) **Leakage Prevention:** Real-world API deployments mandate rigorous time-based cutoff mechanisms to explicitly prevent late-stage CRM duration knowledge from artificially inflating early prediction horizons.
3) **Future Architectures:** Future iterations will experiment with zero-shot sequence inferences via Large Language Models (LLMs) through native API calls [3], and introduce large multi-industry deployment datasets to improve generalization [4], [5].

## VIII. Conclusion

This project successfully designed and benchmarked a Predictive Sales Analytics Engine. By conducting a systematic ablation study, we concluded that while tabular CRM constraints drive the baseline predictive capabilities in B2B SaaS deal forecasting, fusing these constraints with PCA-compressed conversational embeddings via a neural Hybrid architecture yields state-of-the-art predictive performance ($F1 = 0.9710$). This modular system establishes a rigid foundational blueprint for mathematically driven organizational forecasting and lays the groundwork for future multi-modal, LLM-augmented revenue intelligence systems.

## References


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